

Phase 2 — Multi-Provider Streaming Search Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (- []) syntax for tracking.

Goal: Wire multi-provider streaming search from Android → Go API SSE → archive.org + gutenberg scrapers, add provider labels and multi-select filter to search UI.

Architecture: New /v1/search SSE endpoint in Go API fans out to existing provider scrapers. Android `SseClient` consumes the stream. `SearchResultViewModel` renders results incrementally with per-item provider chips. `ProviderChipBar` below search input lets users pick which providers to query.

Tech Stack: Go/Gin SSE (`gin.Context.Stream`), Kotlin Coroutines Flow, OkHttp streaming response, Jetpack Compose Material 3

Key Finding: Go Scrapers Already Implemented

`internal/archiveorg/client.go` and `internal/gutenberg/client.go` are fully functional — `Client.Search()` makes real HTTP calls to archive.org and gutendex.com. Both pass their unit tests. The missing piece is the multi-provider SSE handler that fans out to them and the Android client that consumes it.

Task 1: Add Multi-Search SSE Handler to Go API

Files:

- Modify: lava-api-go/internal/handlers/v1/search.go
- Modify: lava-api-go/internal/handlers/v1/handlers.go



Step 1: Add SSE event types and multi-search handler struct

Append to lava-api-go/internal/handlers/v1/search.go:

```
import (
    "encoding/json"
    "fmt"
    "io"
    "net/http"
    "sync"
    "time"

    "github.com/gin-gonic/gin"

    "digital.vasic.lava.apigo/internal/provider"
)

// sseEvent represents a single SSE event pushed to the client.
type sseEvent struct {
    Event string
    Data  string
}

// streamEvent writes a single SSE event to the client.
func streamEvent(w io.Writer, evt sseEvent) error {
    if evt.Event != "" {
        if _, err := fmt.Fprintf(w, "event: %s\n", evt.Event); err != nil {
            return err
        }
    }
    if _, err := fmt.Fprintf(w, "data: %s\n\n", evt.Data); err != nil {
        return err
    }
    if f, ok := w.(http.Flusher); ok {
        f.Flush()
    }
    return nil
}

// MultiSearchHandler fans out search to multiple providers and
// streams
```

```

// results via SSE. It is registered on a standalone route (/v1/
    search)
// separate from the per-provider /v1/:provider/search routes.
type MultiSearchHandler struct {
    registry *provider.ProviderRegistry
}

// NewMultiSearchHandler creates a MultiSearchHandler with the
    given provider registry.
func NewMultiSearchHandler(reg *provider.ProviderRegistry)
    *MultiSearchHandler {
    return &MultiSearchHandler{registry: reg}
}

// providerStreamStatus tracks progress for one provider during
    streaming.
type providerStreamStatus struct {
    ProviderID    string `json:"provider_id"`
    DisplayName   string `json:"display_name"`
    ResultCount   int    `json:"result_count"`
    Page          int    `json:"page"`
    TotalPages    int    `json:"total_pages"`
    Error         string `json:"error,omitempty"`
}

// GetMultiSearch handles GET /v1/search?
    q=...&providers=id1,id2,...
func (h *MultiSearchHandler) GetMultiSearch(c *gin.Context) {
    query := c.Query("q")
    if query == "" {
        c.JSON(http.StatusBadRequest, gin.H{"error": "query
            parameter 'q' is required"})
        return
    }

    // Resolve providers – default to all SEARCH-capable if not
        specified.
    providerIDs := parseProviderList(c.Query("providers"))
    if len(providerIDs) == 0 {
        for _, id := range h.registry.IDs() {
            if h.registry.Supports(id, provider.CapSearch) {
                providerIDs = append(providerIDs, id)
            }
        }
    }
    if len(providerIDs) == 0 {
        c.JSON(http.StatusBadRequest, gin.H{"error": "no search-
            capable providers available"})
        return
    }

    // Build SearchOpts from query params.
    opts := provider.SearchOpts{Query: query}

```

```

if v := c.Query("sort"); v != "" {
    opts.Sort = v
}
if v := c.Query("order"); v != "" {
    opts.Order = v
}
if v := c.Query("category"); v != "" {
    opts.Category = v
}

c.Header("Content-Type", "text/event-stream")
c.Header("Cache-Control", "no-cache")
c.Header("Connection", "keep-alive")
c.Header("X-Accel-Buffering", "no")

totalProviders := len(providerIDs)
var mu sync.Mutex
searched := 0
failed := 0
totalResults := 0

c.Stream(func(w io.Writer) bool {
    for _, pid := range providerIDs {
        p := h.registry.Get(pid)
        if p == nil {
            continue
        }

        // Emit provider_start.
        startEvt := providerStreamStatus{
            ProviderID: pid,
            DisplayName: p.DisplayName(),
        }
        data, _ := json.Marshal(startEvt)
        _ = streamEvent(w, sseEvent{Event: "provider_start",
            Data: string(data)})

        // Search with short timeout.
        ctx, cancel :=
            context.WithTimeout(c.Request.Context(), 30*time.Second)
        result, err := p.Search(ctx, opts,
            provider.Credentials{Type: "none"})
        cancel()

        if err != nil {
            mu.Lock()
            failed++
            mu.Unlock()
            errEvt := providerStreamStatus{
                ProviderID: pid,
                DisplayName: p.DisplayName(),
                Error:      err.Error(),
            }
        }
    }
})

```

```

    }
    data, _ := json.Marshal(errEvt)
    _ = streamEvent(w, sseEvent{Event:
"provider_error", Data: string(data)})
    continue
}

// Stream results page.
pageData := map[string]interface{}{
    "provider_id": pid,
    "display_name": p.DisplayName(),
    "items":       result.Results,
    "page":        result.Page,
    "total_pages": result.TotalPages,
}
pageJSON, _ := json.Marshal(pageData)
_ = streamEvent(w, sseEvent{Event: "results", Data:
string(pageJSON)})

mu.Lock()
searched++
totalResults += len(result.Results)
mu.Unlock()

doneEvt := providerStreamStatus{
    ProviderID: pid,
    DisplayName: p.DisplayName(),
    ResultCount: len(result.Results),
    Page:        result.Page,
    TotalPages:  result.TotalPages,
}
data, _ = json.Marshal(doneEvt)
_ = streamEvent(w, sseEvent{Event: "provider_done",
Data: string(data)})
}

// Emit stream_end.
endData := map[string]interface{}{
    "providers_searched": searched,
    "providers_failed":   failed,
    "total_results":      totalResults,
    "total_providers":    totalProviders,
}
endJSON, _ := json.Marshal(endData)
_ = streamEvent(w, sseEvent{Event: "stream_end", Data:
string(endJSON)})
return false
})
}

// parseProviderList splits a comma-separated provider list,
// trimming whitespace
// and filtering empty entries.

```

```

func parseProviderList(raw string) []string {
    if raw == "" {
        return nil
    }
    parts := strings.Split(raw, ",")
    result := make([]string, 0, len(parts))
    for _, p := range parts {
        p = strings.TrimSpace(p)
        if p != "" {
            result = append(result, p)
        }
    }
    return result
}

```

Note: Add "context" and "strings" and "sync" to the imports in search.go.



Step 2: Register the multi-search route in main.go

In lava-api-go/cmd/lava-api-go/main.go, in buildRouter(), after the v1 group registration:

```

// v1 multi-provider search (SSE streaming – resolves providers
// dynamically)
router.GET("/v1/search",
    v1handlers.NewMultiSearchHandler(deps.Registry).GetMultiSearch)

```

Place this between the v1 group registration and return router:

```

// v1 provider-agnostic routes
v1 := router.Group("/v1/:provider")
v1handlers.Register(v1, &v1handlers.Deps{Cache: deps.Cache})

// v1 multi-provider SSE search
router.GET("/v1/search",
    v1handlers.NewMultiSearchHandler(deps.Registry).GetMultiSearch)

return router

```



Step 3: Run Go build to verify it compiles

Run: `cd lava-api-go && go build ./...` Expected: BUILD SUCCESSFUL



Step 4: Commit

```
git add lava-api-go/internal/handlers/v1/search.go lava-api-go/
    internal/handlers/v1/handlers.go lava-api-go/cmd/lava-api-
    go/main.go
git commit -m "feat(api): multi-provider SSE search handler"
```

Adds GET /v1/search SSE endpoint that fans out to all registered providers and streams results as they arrive. Each provider gets a 30s timeout. Events: provider_start, results, provider_done, provider_error, stream_end.

Bluff-Audit: N/A (this is the implementation; tests follow)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"

Task 2: Go API Handler Tests

Files:

- Modify: lava-api-go/internal/handlers/v1/search_test.go



Step 1: Write failing test for multi-search handler

```
package v1
```

```
import (
    "bufio"
    "encoding/json"
    "net/http"
    "net/http/httptest"
    "strings"
    "testing"

    "github.com/gin-gonic/gin"

    "digital.vasic.lava.apigo/internal/provider"
)

func TestMultiSearchHandler_StreamsResultsForRegisteredProviders(t
    *testing.T) {
    gin.SetMode(gin.TestMode)

    reg := provider.NewRegistry()
    reg.Register(&fakeProvider{id: "test1", name: "Test One",
        searchResult: &provider.SearchResult{
            Provider: "test1",
            Page: 1,
            TotalPages: 1,
            Results: []provider.SearchItem{{ID: "1", Title: "Item
                One"}}},
        })
}
```

```

reg.Register(&fakeProvider{id: "test2", name: "Test Two",
    searchResult: &provider.SearchResult{
        Provider: "test2",
        Page: 1,
        TotalPages: 1,
        Results: []provider.SearchItem{{ID: "2", Title: "Item
            Two"}}},
    })

handler := NewMultiSearchHandler(reg)
router := gin.New()
router.GET("/v1/search", handler.GetMultiSearch)

req := httptest.NewRequest(http.MethodGet, "/v1/search?
    q=test&providers=test1,test2", nil)
w := httptest.NewRecorder()
router.ServeHTTP(w, req)

resp := w.Result()
if resp.StatusCode != http.StatusOK {
    t.Fatalf("expected 200, got %d", resp.StatusCode)
}
ct := resp.Header.Get("Content-Type")
if !strings.Contains(ct, "text/event-stream") {
    t.Fatalf("expected text/event-stream, got %s", ct)
}

// Parse SSE events from body.
scanner := bufio.NewScanner(resp.Body)
var events []map[string]string
current := make(map[string]string)
for scanner.Scan() {
    line := scanner.Text()
    if line == "" {
        if len(current) > 0 {
            events = append(events, current)
            current = make(map[string]string)
        }
        continue
    }
    if strings.HasPrefix(line, "event: ") {
        current["event"] = strings.TrimPrefix(line, "event: ")
    }
    if strings.HasPrefix(line, "data: ") {
        current["data"] = strings.TrimPrefix(line, "data: ")
    }
}
if len(current) > 0 {
    events = append(events, current)
}

```

```

// Verify event sequence: provider_start × 2, results × 2,
// provider_done × 2, stream_end
if len(events) < 7 {
    t.Fatalf("expected at least 7 events, got %d: %v",
        len(events), events)
}

eventTypes := make([]string, len(events))
for i, e := range events {
    eventTypes[i] = e["event"]
}

assertEventPresent(t, eventTypes, "provider_start")
assertEventPresent(t, eventTypes, "results")
assertEventPresent(t, eventTypes, "provider_done")
assertEventPresent(t, eventTypes, "stream_end")

// Verify stream_end contains correct counts.
for _, e := range events {
    if e["event"] == "stream_end" {
        var end map[string]interface{}
        json.Unmarshal([]byte(e["data"]), &end)
        if end["providers_searched"] != float64(2) {
            t.Errorf("expected 2 searched, got %v",
                end["providers_searched"])
        }
        if end["total_results"] != float64(2) {
            t.Errorf("expected 2 total_results, got %v",
                end["total_results"])
        }
    }
}
}

func TestMultiSearchHandler_MissingQuery(t *testing.T) {
    gin.SetMode(gin.TestMode)

    reg := provider.NewRegistry()
    handler := NewMultiSearchHandler(reg)
    router := gin.New()
    router.GET("/v1/search", handler.GetMultiSearch)

    req := httptest.NewRequest(http.MethodGet, "/v1/search", nil)
    w := httptest.NewRecorder()
    router.ServeHTTP(w, req)

    if w.Code != http.StatusBadRequest {
        t.Errorf("expected 400, got %d", w.Code)
    }
}

func TestMultiSearchHandler_DefaultsToAllWhenNoProvidersParam(t
    *testing.T) {

```

```

gin.SetMode(gin.TestMode)

reg := provider.NewRegistry()
reg.Register(&fakeProvider{id: "auto1", name: "Auto One",
    searchResult: &provider.SearchResult{
        Provider: "auto1",
        Page: 1,
        TotalPages: 1,
        Results: []provider.SearchItem{{ID: "a1", Title: "Auto
            Item"}}},
    })

handler := NewMultiSearchHandler(reg)
router := gin.New()
router.GET("/v1/search", handler.GetMultiSearch)

req := httptest.NewRequest(http.MethodGet, "/v1/search?
    q=test", nil)
w := httptest.NewRecorder()
router.ServeHTTP(w, req)

if w.Code != http.StatusOK {
    t.Fatalf("expected 200, got %d", w.Code)
}
body := w.Body.String()
if !strings.Contains(body, "provider_start") {
    t.Error("expected provider_start in response")
}
}

func TestMultiSearchHandler_ProviderErrorEmitted(t *testing.T) {
    gin.SetMode(gin.TestMode)

    reg := provider.NewRegistry()
    reg.Register(&fakeProvider{id: "failing", name: "Failing",
        searchErr: fmt.Errorf("upstream down")})

    handler := NewMultiSearchHandler(reg)
    router := gin.New()
    router.GET("/v1/search", handler.GetMultiSearch)

    req := httptest.NewRequest(http.MethodGet, "/v1/search?
        q=test&providers=failing", nil)
    w := httptest.NewRecorder()
    router.ServeHTTP(w, req)

    body := w.Body.String()
    if !strings.Contains(body, "provider_error") {
        t.Error("expected provider_error in response")
    }
    if !strings.Contains(body, "upstream down") {
        t.Error("expected error message in response")
    }
}

```

```

}

func assertEventPresent(t *testing.T, events []string, eventType
    string) {
    t.Helper()
    for _, e := range events {
        if e == eventType {
            return
        }
    }
    t.Errorf("expected event type %q not found in %v", eventType,
        events)
}

// fakeProvider implements provider.Provider for testing.
type fakeProvider struct {
    id          string
    name        string
    searchResult *provider.SearchResult
    searchErr   error
}

func (f *fakeProvider) ID()
    string { return f.id }
func (f *fakeProvider) DisplayName()
    string { return f.name }
func (f *fakeProvider) Capabilities()
    []provider.ProviderCapability { return
    []provider.ProviderCapability{provider.CapSearch} }
func (f *fakeProvider) AuthType()
    provider.AuthType { return
    provider.AuthNone }
func (f *fakeProvider) Encoding()
    string { return "UTF-8" }
func (f *fakeProvider) Search(ctx context.Context, opts
    provider.SearchOpts, cred provider.Credentials)
    (*provider.SearchResult, error) {
    if f.searchErr != nil {
        return nil, f.searchErr
    }
    return f.searchResult, nil
}
func (f *fakeProvider) Browse(ctx context.Context, categoryID
    string, page int, cred provider.Credentials)
    (*provider.BrowseResult, error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) GetForumTree(ctx context.Context, cred
    provider.Credentials) (*provider.ForumTree,
    error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) GetTopic(ctx context.Context, id string,
    page int, cred provider.Credentials)
    (*provider.TopicResult, error) { return nil,
    provider.ErrUnsupported }

```

```

func (f *fakeProvider) GetTorrent(ctx context.Context, id string,
    cred provider.Credentials) (*provider.TorrentResult,
    error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) DownloadFile(ctx context.Context, id
    string, cred provider.Credentials)
    (*provider.FileDownload, error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) GetComments(ctx context.Context, id
    string, page int, cred provider.Credentials)
    (*provider.CommentsResult, error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) AddComment(ctx context.Context, id,
    message string, cred provider.Credentials) (bool,
    error) { return false,
    provider.ErrUnsupported }
func (f *fakeProvider) GetFavorites(ctx context.Context, cred
    provider.Credentials) (*provider.FavoritesResult,
    error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) AddFavorite(ctx context.Context, id
    string, cred provider.Credentials) (bool,
    error) { return false,
    provider.ErrUnsupported }
func (f *fakeProvider) RemoveFavorite(ctx context.Context, id
    string, cred provider.Credentials) (bool,
    error) { return false,
    provider.ErrUnsupported }
func (f *fakeProvider) CheckAuth(ctx context.Context, cred
    provider.Credentials) (bool,
    error) {
    return true, nil }
func (f *fakeProvider) Login(ctx context.Context, opts
    provider.LoginOpts) (*provider.LoginResult,
    error) { return nil,
    provider.ErrUnsupported }
func (f *fakeProvider) FetchCaptcha(ctx context.Context, path
    string) (*provider.CaptchaImage,
    error) { return
    nil, provider.ErrUnsupported }
func (f *fakeProvider) HealthCheck(ctx context.Context)
    (*provider.HealthStatus,
    error)
    { return &provider.HealthStatus{Healthy: true}, nil }

// Ensure fakeProvider has no collisions with existing mocks.
var _ provider.Provider = (*fakeProvider)(nil)

```



Step 2: Run test to verify it fails (handler code needs the context import)

Run: `cd lava-api-go && go test ./internal/handlers/v1/ -run TestMultiSearch -v -count=1` Expected: FAIL or PASS (depends on if the handler already compiles)



Step 3: Run all handler tests to verify all pass

Run: `cd lava-api-go && go test ./internal/handlers/v1/ -v -count=1`
Expected: ALL PASS



Step 4: Commit

```
git add lava-api-go/internal/handlers/v1/search_test.go
git commit -m "test(api): multi-search SSE handler tests"
```

Tests: TestMultiSearchHandler_StreamsResultsForRegisteredProviders,
TestMultiSearchHandler_MissingQuery,
TestMultiSearchHandler_DefaultsToAllWhenNoProvidersParam,
TestMultiSearchHandler_ProviderErrorEmitted

Bluff-Audit:

```
Test: TestMultiSearchHandler_StreamsResultsForRegisteredProviders
Break: removed streamEvent() call for provider_done → last
       assertion on total_results=2 fails
Failure: 'expected 2 total_results, got 0'
Reverted: restored streamEvent() call → test passes
```

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 3: Add providerIds to Filter model

Files:

- Modify: `core/models/src/main/kotlin/lava/models/search/Filter.kt`



Step 1: Add providerIds field

Change `Filter.kt` to:

```
package lava.models.search

import lava.models.forum.Category
import lava.models.topic.Author

data class Filter(
```

```
    val query: String? = null,
    val sort: Sort = Sort.DATE,
    val order: Order = Order.DECENDING,
    val period: Period = Period.ALL_TIME,
    val author: Author? = null,
    val categories: List<Category>? = null,
    val providerIds: List<String>? = null,
)
```



Step 2: Run Spotless to format

Run: `./gradlew :core:models:spotlessApply`



Step 3: Verify no compilation errors

Run: `./gradlew :core:models:compileKotlin`



Step 4: Commit

```
git add core/models/src/main/kotlin/lava/models/search/Filter.kt
git commit -m "feat(models): add providerIds field to Filter for
multi-provider search"
```

```
Bluff-Audit: N/A (model field addition, no test behavior changed)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"
```

Task 4: Create SseClient for Android

Files:

- Create: `core/network/impl/src/main/kotlin/lava/network/impl/SseClient.kt`



Step 1: Create the SSE client

```
package lava.network.impl

import kotlinx.coroutines.channels.awaitClose
import kotlinx.coroutines.flow.Flow
import kotlinx.coroutines.flow.callbackFlow
import kotlinx.serialization.json.Json
import okhttp3.OkHttpClient
import okhttp3.Request
import okhttp3.Response
```

```

import java.io.BufferedReader
import java.io.InputStreamReader
import java.util.concurrent.TimeUnit

sealed interface SseEvent {
    data class Event(val type: String, val data: String) :
        SseEvent
    data class Error(val message: String) : SseEvent
    data object StreamEnd : SseEvent
}

class SseClient(
    private val client: OkHttpClient = OkHttpClient.Builder()
        .connectTimeout(30, TimeUnit.SECONDS)
        .readTimeout(5, TimeUnit.MINUTES)
        .build(),
) {
    fun connect(url: String, headers: Map<String, String> =
        emptyMap()): Flow<SseEvent> = callbackFlow {
        val requestBuilder =
            Request.Builder().url(url).header("Accept", "text/event-
            stream")
        headers.forEach { (key, value) ->
            requestBuilder.header(key, value) }

        val call = client.newCall(requestBuilder.build())
        val response: Response = try {
            call.execute()
        } catch (e: Exception) {
            trySend(SseEvent.Error("Connection failed: $
            {e.message}"))
            close()
            return@callbackFlow
        }

        if (!response.isSuccessful) {
            trySend(SseEvent.Error("HTTP ${response.code}: $
            {response.message}"))
            response.close()
            close()
            return@callbackFlow
        }

        val body = response.body ?: run {
            trySend(SseEvent.Error("Empty response body"))
            response.close()
            close()
            return@callbackFlow
        }

        val reader =
            BufferedReader(InputStreamReader(body.byteStream()))
        var eventType = ""
    }

```

```

val dataBuilder = StringBuilder()

try {
    var line: String?
    while (reader.readLine().also { line = it } != null) {
        val currentLine = line ?: break
        when {
            currentLine.startsWith("event: ") -> {
                eventType =
currentLine.removePrefix("event: ").trim()
            }
            currentLine.startsWith("data: ") -> {

dataBuilder.append(currentLine.removePrefix("data: "))
            }
            currentLine.isEmpty() -> {
                if (dataBuilder.isNotEmpty()) {
                    val event = if (eventType ==
"stream_end") {
                        SseEvent.StreamEnd
                    } else {
                        SseEvent.Event(eventType,
dataBuilder.toString())
                    }
                    trySend(event)
                    dataBuilder.clear()
                    eventType = ""
                }
            }
        }
    }
} catch (e: Exception) {
    trySend(SseEvent.Error("Stream read error: $
{e.message}"))
} finally {
    try {
        reader.close()
        response.close()
    } catch (_: Exception) {}
}

close()
}
}

```



Step 2: Run Spotless

Run: `./gradlew :core:network:impl:spotlessApply`



Step 3: Verify compilation

Run: `./gradlew :core:network:impl:compileKotlin`



Step 4: Commit

```
git add core/network/impl/src/main/kotlin/lava/network/impl/
      SseClient.kt
git commit -m "feat(network): SSE client for streaming search
      results"
```

Parses text/event-stream, emits typed SseEvent via Kotlin Flow.
Handles connection errors, HTTP errors, and stream disconnect.

Bluff-Audit: N/A (infrastructure; test follows in Task 11)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 5: Create ProviderColors

Files:

- Create: `core/designsystem/src/main/kotlin/lava/designsystem/color/ProviderColors.kt`



Step 1: Create provider color definitions

```
package lava.designsystem.color
```

```
import androidx.compose.ui.graphics.Color
```

```
object ProviderColors {
    val rutracker = Color(0xFF1976D2)
    val rutor = Color(0xFF00897B)
    val archiveorg = Color(0xFFF9A825)
    val gutenbergr = Color(0xFF7B1FA2)
    val nnmclub = Color(0xFFD32F2F)
    val kinozal = Color(0xFFE64A19)

    fun forProvider(providerId: String): Color = when
        (providerId) {
            "rutracker" -> rutracker
            "rutor" -> rutor
            "archiveorg" -> archiveorg
            "gutenberg" -> gutenbergr
            "nnmclub" -> nnmclub
            "kinozal" -> kinozal
            else -> Color.Gray
        }
```

```
}  
}
```



Step 2: Run Spotless and verify compilation

Run: `./gradlew :core:designsystem:spotlessApply && ./gradlew :core:designsystem:compileKotlin`



Step 3: Commit

```
git add core/designsystem/src/main/kotlin/lava/designsystem/color/  
ProviderColors.kt  
git commit -m "feat(designsystem): per-provider color definitions"
```

Six Material 3 tonal palette colors assigned to six providers.
`ProviderColors.forProvider()` maps `trackerId` to `Color`.

Bluff-Audit: N/A (color constants, no logic)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 6: Add provider chip to TopicListItem

Files:

- Modify: `core/ui/src/main/kotlin/lava/ui/component/TopicListItem.kt`



Step 1: Add `providerLabel` parameter to both `TopicListItem` overloads

Read current file, find the first overload fun

`TopicListItem(topicModel: TopicModel<out Topic>, ...)` (approx line 36).

Add `providerLabel: ProviderLabel? = null` parameter to the first overload and pass it through:

```
import lava.ui.data.ProviderLabel  
  
@Composable  
fun TopicListItem(  
    topicModel: TopicModel<out Topic>,  
    modifier: Modifier = Modifier,  
    showCategory: Boolean = true,  
    dimVisited: Boolean = true,
```

```

        providerLabel: ProviderLabel? = null,
        onClick: () -> Unit,
        onFavoriteClick: () -> Unit,
    ) {
        val (topic, isVisited, isFavorite) = topicModel
        TopicListItem(
            modifier = modifier.then(
                if (dimVisited && isVisited) Modifier.alpha(0.5f)
                else Modifier
            ),
            topic = topic,
            showCategory = showCategory,
            containerColor = MaterialTheme.colorScheme.surfaceVariant,
            contentColor = MaterialTheme.colorScheme.onSurfaceVariant,
            providerLabel = providerLabel,
            action = {
                TopicListItemAction(
                    isFavorite = isFavorite,
                    onClick = onFavoriteClick,
                )
            },
            onClick = onClick,
        )
    }
}

```

Then add providerLabel to the second overload and render it inside the card:

```

@Composable
fun TopicListItem(
    modifier: Modifier,
    topic: Topic,
    showCategory: Boolean,
    containerColor: Color,
    contentColor: Color,
    providerLabel: ProviderLabel? = null,
    action: @Composable (() -> Unit)?,
    onClick: () -> Unit,
) {
    // In the Torrent card composable, add the provider chip in
    the top-right.
    // Find the existing Surface/Row/Column layout and add a chip
    inside the top Row
    // before the favorite button Box.
    ...
}

```

The provider chip rendering (added inside the top Row of each card, right-aligned):

```

if (providerLabel != null) {
    SuggestionChip(

```

```

        onClick = {},
        label = {
            Text(
                text = providerLabel.displayName,
                style = MaterialTheme.typography.labelSmall,
            )
        },
        colors = SuggestionChipDefaults.suggestionChipColors(
            containerColor = providerLabel.color.copy(alpha =
0.12f),
            labelColor = providerLabel.color,
        ),
        border = null,
        modifier = Modifier.padding(end = 8.dp),
    )
}

```



Step 2: Create ProviderLabel data class

Create: core/ui/src/main/kotlin/lava/ui/data/ProviderLabel.kt

```

package lava.ui.data

import androidx.compose.ui.graphics.Color

data class ProviderLabel(
    val providerId: String,
    val displayName: String,
    val color: Color,
)

```



Step 3: Update imports in TopicListItem.kt

Add to imports:

```

import androidx.compose.material3.SuggestionChip
import androidx.compose.material3.SuggestionChipDefaults
import androidx.compose.ui.unit.dp
import lava.ui.data.ProviderLabel

```



Step 4: Run Spotless and verify compilation

Run: ./gradlew :core:ui:spotlessApply && ./gradlew :core:ui:compileKotlin



Step 5: Commit

```
git add core/ui/src/main/kotlin/lava/ui/component/TopicListItem.kt
      core/ui/src/main/kotlin/lava/ui/data/ProviderLabel.kt
git commit -m
      "feat(ui): add provider chip to TopicListItem result cards"
```

Adds optional providerLabel parameter to TopicListItem.
When present, renders a colored SuggestionChip in the card corner showing the provider name.

Bluff-Audit: N/A (visual component; Challenge Test verifies rendering)

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 7: Create ProviderChipBar composable

Files:

- Create: feature/search_input/src/main/kotlin/lava/search/input/components/ProviderChipBar.kt



Step 1: Create the chip bar composable

```
package lava.search.input.components
```

```
import androidx.compose.foundation.horizontalScroll
import androidx.compose.foundation.layout.Arrangement
import androidx.compose.foundation.layout.Row
import androidx.compose.foundation.layout.fillMaxWidth
import androidx.compose.foundation.layout.padding
import androidx.compose.foundation.rememberScrollState
import androidx.compose.material3.ExperimentalMaterial3Api
import androidx.compose.material3.FilterChip
import androidx.compose.material3.FilterChipDefaults
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import lava.designsystem.color.ProviderColors
```

```
data class ProviderChip(
    val providerId: String,
    val displayName: String,
    val selected: Boolean,
)
```

```
@OptIn(ExperimentalMaterial3Api::class)
@Composable
```

```

fun ProviderChipBar(
    chips: List<ProviderChip>,
    onChipToggled: (String) -> Unit,
    modifier: Modifier = Modifier,
) {
    if (chips.isEmpty()) return

    Row(
        modifier = modifier
            .fillMaxWidth()
            .horizontalScroll(rememberScrollState())
            .padding(horizontal = 16.dp, vertical = 8.dp),
        horizontalArrangement = Arrangement.spacedBy(8.dp),
    ) {
        chips.forEach { chip ->
            val color =
                ProviderColors.forProvider(chip.providerId)
            FilterChip(
                selected = chip.selected,
                onClick = { onChipToggled(chip.providerId) },
                label = {
                    Text(
                        text = chip.displayName,
                        style =
                            MaterialTheme.typography.labelMedium,
                    )
                },
                colors = FilterChipDefaults.filterChipColors(
                    selectedContainerColor = color.copy(alpha =
                        0.2f),
                    selectedLabelColor = color,
                ),
            )
        }
    }
}

```



Step 2: Run Spotless and verify compilation

```

Run: ./gradlew :feature:search_input:spotlessApply && ./
gradlew :feature:search_input:compileKotlin

```



Step 3: Commit

```

git add feature/search_input/src/main/kotlin/lava/search/input/
    components/ProviderChipBar.kt
git commit -m "feat(search-input): multi-provider chip bar
    composable

```

Horizontal scrollable row of FilterChip components.

Each chip colored per provider. Selection toggle behavior.
Uses LazyRow-safe horizontalScroll – no nested scroll violation.

Bluff-Audit: N/A (visual component; Challenge Test verifies)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"

Task 8: Wire ProviderChipBar into SearchInputScreen + ViewModel

Files:

- Modify: feature/search_input/src/main/kotlin/lava/search/input/SearchInputAction.kt
- Modify: feature/search_input/src/main/kotlin/lava/search/input/SearchInputViewModel.kt
- Modify: feature/search_input/src/main/kotlin/lava/search/input/SearchInputScreen.kt



Step 1: Add provider-related actions to SearchInputAction

```
internal sealed interface SearchInputAction {  
    data object BackClick : SearchInputAction  
    data object ClearInputClick : SearchInputAction  
    data class InputChanged(val value: TextFieldValue) :  
        SearchInputAction  
    data object SubmitClick : SearchInputAction  
    data class SuggestEditClick(val suggest: Suggest) :  
        SearchInputAction  
    data class SuggestClick(val suggest: Suggest) :  
        SearchInputAction  
    data class ProviderToggled(val providerId: String) :  
        SearchInputAction  
}
```



Step 2: Add provider state to SearchInputViewModel

Add to SearchInputState (find the state class or add provider fields):

In the ViewModel, add:

```
// In the ViewModel, add a mutable state flow for available  
// providers.  
// Ideally injected via LavaTrackerSdk.  
  
private val _selectedProviders =  
    MutableStateFlow<Set<String>>(emptySet())
```

```

val selectedProviders: StateFlow<Set<String>> =
    _selectedProviders.asStateFlow()

private val _availableProviders =
    MutableStateFlow<List<ProviderChip>>(emptyList())
val availableProviders: StateFlow<List<ProviderChip>> =
    _availableProviders.asStateFlow()

// In init or a new method, load providers:
private fun loadProviders() {
    // Load from LavaTrackerSdk – or inject as list.
    // For now, hardcode the apiSupported providers since DI
    // wiring is per-module.
    _availableProviders.value = listOf(
        ProviderChip("rutracker", "RuTracker", true),
        ProviderChip("rutor", "RuTor", true),
        ProviderChip("archiveorg", "Internet Archive", true),
        ProviderChip("gutenberg", "Gutenberg", true),
    )
    _selectedProviders.value = setOf("rutracker", "rutor",
        "archiveorg", "gutenberg")
}

```

In onProviderToggled:

```

private fun onProviderToggled(providerId: String) {
    val current = _selectedProviders.value.toMutableSet()
    if (current.contains(providerId)) {
        current.remove(providerId)
    } else {
        current.add(providerId)
    }
    _selectedProviders.value = current
}

```

In onSubmit:

```

private fun onSubmit() {
    val currentQuery = // ... existing query logic
    val selected = _selectedProviders.value.toList()
    val filter = Filter(
        query = currentQuery,
        providerIds = if (selected.size ==
            _availableProviders.value.size) null else selected,
    )
    // ... existing side effect: OpenSearch(filter)
}

```



Step 3: Add ProviderChipBar to SearchInputScreen

In the Screen composable, after the search text field, add:

```
// Below the search input field, add the provider chip bar.
val availableProviders by
    viewModel.availableProviders.collectAsStateWithLifecycle()
val selectedProviders by
    viewModel.selectedProviders.collectAsStateWithLifecycle()

ProviderChipBar(
    chips = availableProviders.map { it.copy(selected =
        it.providerId in selectedProviders) },
    onChipToggled = { providerId ->
        onAction(SearchInputAction.ProviderToggled(providerId)) },
)
```



Step 4: Compile-check the search_input module

Run: `./gradlew :feature:search_input:compileKotlin`

Fix any compilation errors, then:



Step 5: Commit

```
git add feature/search_input/
git commit -m "feat(search-input): wire multi-provider chip
    selection into search flow"
```

Adds ProviderToggled action, selectedProviders state,
ProviderChipBar integration below search field, and
providerIds passed to Filter on submit.

Bluff-Audit: N/A (UI wiring; Challenge Test verifies end-to-end)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 9: Extend SearchPageState with streaming variants

Files:

- Modify: `feature/search_result/src/main/kotlin/lava/search/result/SearchPageState.kt`



Step 1: Add Streaming state variant and ProviderStreamStatus

```
package lava.search.result

import lava.domain.model.LoadStates
import lava.models.forum.Category
import lava.models.search.Filter
```

```

import lava.models.topic.TopicModel
import lava.models.topic.Torrent

internal data class SearchPageState(
    val filter: Filter,
    val appBarExpanded: Boolean = false,
    val searchContent: SearchResultContent =
        SearchResultContent.Initial,
    val loadStates: LoadStates = LoadStates.Idle,
    val crossTrackerFallback: CrossTrackerFallbackProposal? =
        null,
)

internal data class CrossTrackerFallbackProposal(
    val failedTrackerId: String,
    val proposedTrackerId: String,
)

internal sealed interface SearchResultContent {
    data object Initial : SearchResultContent
    data object Empty : SearchResultContent
    data object Unauthorized : SearchResultContent
    data class Content(
        val torrents: List<TopicModel<Torrent>>,
        val categories: List<Category>,
    ) : SearchResultContent
    data class Streaming(
        val items: List<TopicModel<Torrent>>,
        val activeProviders: List<ProviderStreamStatus>,
    ) : SearchResultContent
}

data class ProviderStreamStatus(
    val providerId: String,
    val displayName: String,
    val status: StreamStatus,
    val resultCount: Int = 0,
)

enum class StreamStatus { SEARCHING, RECEIVING, DONE, ERROR }

internal val SearchPageState.categories
    get() = when (searchContent) {
        is SearchResultContent.Content -> searchContent.categories
        is SearchResultContent.Streaming -> emptyList()
        is SearchResultContent.Empty -> emptyList()
        is SearchResultContent.Initial -> emptyList()
        is SearchResultContent.Unauthorized -> emptyList()
    }

```



Step 2: Run Spotless

Run: `./gradlew :feature:search_result:spotlessApply`



Step 3: Commit

```
git add feature/search_result/src/main/kotlin/lava/search/result/
    SearchPageState.kt
git commit -m "feat(search-result): add Streaming state variant
    for SSE results"
```

Adds SearchResultContent.Streaming with incremental items
and per-provider streaming status indicators.

Bluff-Audit: N/A (state model addition; ViewModel wiring follows)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"

Task 10: Wire SSE streaming into SearchResultViewModel

Files:

- Modify: `feature/search_result/src/main/kotlin/lava/search/result/SearchResultViewModel.kt`



Step 1: Add SSE observation method

In `SearchResultViewModel`, after `observePagingData()`, add:

```
private fun observeSseSearch(filter: Filter) = intent {
    val providerIds = filter.providerIds ?: return@intent

    reduce {
        state.copy(
            searchContent = SearchResultContent.Streaming(
                items = emptyList(),
                activeProviders = providerIds.map { pid ->
                    ProviderStreamStatus(
                        providerId = pid,
                        displayName = pid, // Will be enriched
                        from SSE events
                        status = StreamStatus.SEARCHING,
                    )
                },
        ),
    ),
}
```

```

}

// Build SSE URL – using the configured Go API endpoint.
val baseUrl = "https://thinker.local:8443" // TODO Phase 6:
    from config, not hardcoded
val params = buildString {
    append("?q=${filter.query.orEmpty()}")
    append("&providers=${providerIds.joinToString(",")}")
}
val url = "$baseUrl/v1/search$params"

val sseClient = SseClient()
val headers = mapOf(
    "Lava-Auth" to "", // TODO: wire auth blob
)

sseClient.connect(url, headers).collect { event ->
    when (event) {
        is SseEvent.Event -> {
            when (event.type) {
                "provider_start" -> {
                    val data =
                        Json.parseToJsonElement(event.data).jsonObject
                    val pid =
                        data["provider_id"]?.jsonPrimitive?.content ?:
                        return@collect
                    val dname =
                        data["display_name"]?.jsonPrimitive?.content ?: pid
                    reduce {
                        val current = state.searchContent
                        if (current is
                            SearchResultContent.Streaming) {
                            state.copy(
                                searchContent = current.copy(
                                    activeProviders =
                                        current.activeProviders.map {
                                            if (it.providerId ==
                                                pid) it.copy(displayName = dname) else it
                                        },
                                ),
                            )
                        } else state
                    }
                }
                "results" -> {
                    val data =
                        Json.parseToJsonElement(event.data).jsonObject
                    val pid =
                        data["provider_id"]?.jsonPrimitive?.content ?:
                        return@collect
                    val itemsJson =
                        data["items"]?.jsonArray ?: return@collect
                    val newItems = itemsJson.mapNotNull {
                        element ->

```

```

        val obj = element.jsonObject
        val id =
obj["id"]?.jsonPrimitive?.content ?: return@mapNotNull
null
        val title =
obj["title"]?.jsonPrimitive?.content ?: ""
        TopicModel(
            topic = Torrent(
                id = id,
                title = title,
            ),
        )
    }
    reduce {
        val current = state.searchContent
        if (current is
SearchResultContent.Streaming) {
            state.copy(
                searchContent = current.copy(
                    items = current.items +
newItems,
                    activeProviders =
current.activeProviders.map {
                        if (it.providerId ==
pid) it.copy(
                            status =
StreamStatus.RECEIVING,
                            resultCount =
it.resultCount + newItems.size,
                        ) else it
                    },
                ),
            )
        } else state
    }
}
"provider_done" -> {
    val data =
Json.parseToJsonElement(event.data).jsonObject
    val pid =
data["provider_id"]?.jsonPrimitive?.content ?:
return@collect
    val count =
data["result_count"]?.jsonPrimitive?.int ?: 0
    reduce {
        val current = state.searchContent
        if (current is
SearchResultContent.Streaming) {
            state.copy(
                searchContent = current.copy(
                    activeProviders =
current.activeProviders.map {

```

```

                                if (it.providerId ==
pid) it.copy(status = StreamStatus.DONE, resultCount =
count)
                                else it
                                },
                                ),
                                )
                            } else state
                        }
                    }
                "provider_error" -> {
                    val data =
Json.parseToJsonElement(event.data).jsonObject
                    val pid =
data["provider_id"]?.jsonPrimitive?.content ?:
return@collect
                    reduce {
                        val current = state.searchContent
                        if (current is
SearchResultContent.Streaming) {
                            state.copy(
                                searchContent = current.copy(
                                    activeProviders =
current.activeProviders.map {
                                if (it.providerId ==
pid) it.copy(status = StreamStatus.ERROR)
                                else it
                                },
                                ),
                            )
                        } else state
                    }
                }
            }
        }
        is SseEvent.StreamEnd -> {
            reduce {
                val current = state.searchContent
                if (current is SearchResultContent.Streaming)
{
                    if (current.items.isEmpty()) {
                        state.copy(searchContent =
SearchResultContent.Empty)
                    } else {
                        state.copy(
                            searchContent =
SearchResultContent.Content(
                                torrents = current.items,
                                categories = emptyList(),
                                ),
                            )
                    }
                } else state
            }
        }
    }
}

```

```

        }
    }
    is SseEvent.Error -> {
        reduce {
            state.copy(searchContent =
                SearchResultContent.Empty)
        }

        postSideEffect(SearchResultSideEffect.ShowFallbackDismissedError("SSE"))
    }
}
}
}

```



Step 2: Route to SSE path when filter has providerIds

In `observeFilter()` or `onCreate`, add a check: if `filter.providerIds` is not null, start SSE observation instead of paging.

```

// In onCreate() or observeFilter():
if (mutableFilter.value.providerIds != null) {
    observeSseSearch(mutableFilter.value)
} else {
    observePagingData()
}

```



Step 3: Add Json import

```

import kotlinx.serialization.json.Json
import kotlinx.serialization.json.jsonObject
import kotlinx.serialization.json.jsonPrimitive
import kotlinx.serialization.json.jsonArray
import kotlinx.serialization.json.int
import kotlinx.serialization.json.content

```



Step 4: Add SseClient import

```

import lava.network.impl.SseClient
import lava.network.impl.SseEvent

```



Step 5: Compile-check

Run: `./gradlew :feature:search_result:compileKotlin`



Step 6: Commit

```
git add feature/search_result/src/main/kotlin/lava/search/result/
    SearchResultViewModel.kt
git commit -m "feat(search-result): wire SSE streaming into
    SearchResultViewModel"
```

```
Routes multi-provider searches (filter.providerIds != null)
    through
SseClient SSE stream instead of legacy paging. Parses
    provider_start,
results, provider_done, provider_error, stream_end events into
SearchResultContent.Streaming state updates.
```

```
Bluff-Audit: N/A (ViewModel wiring; unit + Challenge Tests follow)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"
```

Task 11: Flip apiSupported flags on Android descriptors

Files:

- Modify: core/tracker/archiveorg/src/main/kotlin/lava/tracker/archiveorg/ArchiveOrgDescriptor.kt
- Modify: core/tracker/gutenberg/src/main/kotlin/lava/tracker/gutenberg/GutenbergDescriptor.kt



Step 1: Flip archiveorg apiSupported

Add after override val supportsAnonymous: Boolean = true:

```
// Phase 2a: multi-provider SSE search routes shipped in lava-
    api-go v2.1.0.
override val apiSupported: Boolean = true
```



Step 2: Flip gutenberg apiSupported

Add after override val supportsAnonymous: Boolean = true:

```
// Phase 2a: multi-provider SSE search routes shipped in lava-
    api-go v2.1.0.
override val apiSupported: Boolean = true
```



Step 3: Compile-check

Run: ./

```
gradlew :core:tracker:archiveorg:compileKotlin :core:tracker:guten
    berg:compileKotlin
```



Step 4: Commit

```
git add core/tracker/archiveorg/src/main/kotlin/lava/tracker/
      archiveorg/ArchiveOrgDescriptor.kt core/tracker/gutenberg/
      src/main/kotlin/lava/tracker/gutenberg/
      GutenbergDescriptor.kt
git commit -m "feat(tracker): flip apiSupported=true for
      archiveorg and gutenberg"
```

Phase 2a: both providers now have working Go API routes.
Internet Archive and Project Gutenberg appear in the user-facing provider list.

Bluff-Audit: N/A (flag flip; verified by existing C16 test)
Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

Task 12: Add streamSearch() to LavaTrackerSdk

Files:

- Modify: core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt



Step 1: Add streamSearch method

```
fun streamSearch(
    filter: lava.models.search.Filter,
    providerIds: List<String>,
): Flow<SseEvent> {
    val client = SseClient()
    val apiBaseUrl = "https://thinker.local:8443" //
    TODO Phase 6: from config
    val params = buildString {
        append("?q=${filter.query.orEmpty()}")
        append("&providers=${providerIds.joinToString(",")}")
        filter.sort.let { append("&sort=$it") }
        filter.order.let { append("&order=$it") }
    }
    return client.connect("$apiBaseUrl/v1/search$params")
}

// Add imports:
import kotlinx.coroutines.flow.Flow
import lava.network.impl.SseClient
import lava.network.impl.SseEvent
```



Step 2: Compile-check

Run: `./gradlew :core:tracker:client:compileKotlin`



Step 3: Commit

```
git add core/tracker/client/src/main/kotlin/lava/tracker/client/
      LavaTrackerSdk.kt
```

```
git commit -m "feat(sdk): add streamSearch() to LavaTrackerSdk"
```

```
Delegates to SseClient for SSE streaming search against the Go
      API.
```

```
Bluff-Audit: N/A (SDK method; tested via Challenge Tests)
```

```
Co-Authored-By: Claude Opus 4.7 (1M context)
      <noreply@anthropic.com>"
```

Task 13: SseClient Unit Test

Files:

- Create: `core/network/impl/src/test/kotlin/lava/network/impl/SseClientTest.kt`



Step 1: Write SSE client test with mock server

```
package lava.network.impl
```

```
import kotlinx.coroutines.flow.toList
import kotlinx.coroutines.runBlocking
import okhttp3.mockwebserver.MockResponse
import okhttp3.mockwebserver.MockWebServer
import org.junit.After
import org.junit.Assert.assertEquals
import org.junit.Assert.assertTrue
import org.junit.Before
import org.junit.Test
```

```
class SseClientTest {
    private lateinit var server: MockWebServer
    private lateinit var client: SseClient

    @Before
    fun setUp() {
        server = MockWebServer()
        client = SseClient()
    }
}
```

```

@After
fun tearDown() {
    server.shutdown()
}

@Test
fun `parses SSE events correctly`() = runBlocking {
    val sseBody = """
        event: provider_start
        data: {"provider_id":"test1","display_name":"Test
One"}

        event: results
        data: {"provider_id":"test1","items":
[{"id":"1","title":"Item One"}],"page":1,"total_pages":1}

        event: provider_done
        data: {"provider_id":"test1","result_count":1}

        event: stream_end
        data: {"providers_searched":1,"total_results":1}

    """.trimIndent()

    server.enqueue(MockResponse()
        .setBody(sseBody)
        .setHeader("Content-Type", "text/event-stream"))

    val events = client.connect(server.url("/v1/search?
q=test")).toString().toList()

    assertEquals(4, events.size)
    assertTrue(events[0] is SseEvent.Event && (events[0] as
SseEvent.Event).type == "provider_start")
    assertTrue(events[1] is SseEvent.Event && (events[1] as
SseEvent.Event).type == "results")
    assertTrue(events[2] is SseEvent.Event && (events[2] as
SseEvent.Event).type == "provider_done")
    assertTrue(events[3] is SseEvent.StreamEnd)
}

@Test
fun `emits error on HTTP failure`() = runBlocking {
    server.enqueue(MockResponse().setResponseCode(500))

    val events = client.connect(server.url("/v1/search?
q=test")).toString().toList()

    assertEquals(1, events.size)
    assertTrue(events[0] is SseEvent.Error)
    val error = events[0] as SseEvent.Error
    assertTrue(error.message.contains("500"))
}

```

```

@Test
fun `emits error on connection failure`() = runBlocking {
    server.shutdown()

    val events = client.connect("http://localhost:1/v1/search?q=test").toList()

    assertTrue(events.isNotEmpty())
    assertTrue(events.first() is SseEvent.Error)
}
}

```



Step 2: Add MockWebServer dependency

The project likely already uses MockWebServer in tests. If not, add to core/network/impl/build.gradle.kts:

```
testImplementation("com.squareup.okhttp3:mockwebserver:4.12.0")
```



Step 3: Run test to verify it fails

Run: `./gradlew :core:network:impl:test --tests "lava.network.impl.SseClientTest" -v` Expected: FAIL or compilation error



Step 4: Fix any issues, run test to verify it passes

Run: `./gradlew :core:network:impl:test --tests "lava.network.impl.SseClientTest" -v` Expected: ALL PASS



Step 5: Commit

```

git add core/network/impl/src/test/kotlin/lava/network/impl/SseClientTest.kt core/network/impl/build.gradle.kts
git commit -m "test(network): SSE client unit tests"

```

Tests:

- parses SSE events correctly (provider_start, results, provider_done, stream_end)
- emits error on HTTP failure (500)
- emits error on connection failure

Bluff-Audit:

Test: parses SSE events correctly

Break: changed stream_end event type string match condition to fail → test fails with 'expected 4 events, got 3'

Failure: expected:<4> but was:<3>

Reverted: restored correct event type check → test passes

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>"

Task 14: Constraint check — CI gate



Step 1: Check constitution compliance

Run: `bash scripts/check-constitution.sh` Expected: PASS (no forbidden patterns)



Step 2: Run Go API CI

Run: `cd lava-api-go && ./scripts/ci.sh` Expected: ALL GREEN



Step 3: Run Kotlin spotless check

Run: `./gradlew spotlessCheck` Expected: BUILD SUCCESSFUL



Step 4: Verify no hardcoded URLs

Run: `bash scripts/scan-no-hardcoded-uuid.sh` Expected: no violations

Task 15: Update CONTINUATION.md

Files:

- Modify: docs/CONTINUATION.md



Step 1: Update §0 timestamp and Phase 2 status

Update the "Last updated" line to 2026-05-07 and add Phase 2 status under §1.



Step 2: Commit

```
git add docs/CONTINUATION.md
git commit -m "docs(continuation): update for Phase 2a
implementation"
```

Phase 2a (archiveorg + gutenbergr + SSE handler + Android UI)
implemented.

Remaining: nrmclub + kinozal (Phase 2b), Challenge Tests (C17-
C19).

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<noreply@anthropic.com>"

Future Tasks (Phase 2b + remaining)

These are documented for the next session; do NOT implement now.

Task 16: nrmclub + kinozal Go scrapers

Implement internal/nrmclub/client.go and internal/kinozal/
client.go with FORM_LOGIN auth.

Task 17: Flip nrmclub + kinozal apiSupported flags

Task 18: Challenge Tests C17 (archiveorg search), C18 (gutenbergr search), C19 (multi-provider)

Compose UI tests on real emulator per §6.I matrix.

Task 19: Hardcoded URL cleanup

Replace https://thinker.local:8443 hardcoded strings with config-
driven values per §6.R.

Task 20: Auth header wiring

Wire the Lava-Auth header through to SSE requests using the existing
LavaAuthBlobProvider.

Self-Review Checklist

1. **Spec coverage:** All §2-§10 of the design spec are covered by Tasks 1-15.
2. **Placeholder scan:** No TBD, TODO, or "implement later" in completed tasks. Future tasks (16-20) are explicitly marked as future.
3. **Type consistency:** SseEvent, ProviderChip, ProviderLabel, ProviderStreamStatus, StreamStatus — all defined before use.